

Intelligent Remote Controlled Antenna Matrix for Shortwave Radio

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Foreword

This bachelor thesis marks the final stage of my studies in Electronics-ICT and represents an important learning experience for me. The project gave me the opportunity to work on a challenging and practical topic that combines RF design, PCB development, measurement techniques, and embedded software.

During this thesis, I learned not only technical skills, but also the importance of making design decisions step by step, based on calculations, testing, and feedback. The project was sometimes challenging, especially since it required balancing technical performance, cost, and practical implementation, but it was also very valuable and motivating.

I would like to sincerely thank my company promoter, Mr. Pedro Wyns, for his guidance, feedback, and practical suggestions throughout the project. I would also like to thank my college promoters for their support and advice during the development of this thesis. In addition, I would like to express my gratitude to everyone who supported me during this period.

Finally, I am grateful for the opportunity to work on this project, which allowed me to deepen my knowledge and gain practical experience in an area that strongly interests me.

Summary

HF radio stations employ several antennas since there is no perfect antenna for all operating frequencies and modes. Switching antennas manually is inconvenient and involves a higher probability of making mistakes. The objective of the bachelor thesis was to design an intelligent remotely controlled antenna matrix which would allow switching between eight antennas and two HF transceivers, and which would include additional functionalities of measuring forward and reflected power, indicating SWR, and detecting operating frequency.

Starting from the system analysis and definition of functional, electrical, and mechanical requirements, a design was developed based on a two-board concept where the RF section was separated from control electronics. The RF board includes the antenna switching matrix, directional couplers, and frequency pickup circuits. The control board includes the CYD-based controller, relay drivers, ADC interface, and the control algorithm. The design includes relay selection, RF layout, grounding, and safety logic. Several revisions of relay selection were necessary due to high power stress in RF section under mismatch conditions.

As a result, an advanced design of HF antenna matrix was developed. It allows antenna switching, forward and reflected power measurement, SWR indication, and frequency detection. In addition, there is an opportunity to implement control functionality by means of remote communication. The project demonstrates that an intelligent antenna matrix can be realized using embedded design technology, considering proper relay selection and RF layout.

Table of contents

1. INTRODUCTION	1
2. PROBLEM STATEMENT	2
2.1. Operational need	2
2.2. Main technical difficulty	2
2.2.1. Relay selection problem	3
2.2.2. Ground-clamp relay problem	4
2.3. Measurements and integration difficulties	5
2.3.1. Forward and reflected power measurements and SWR calculation	5
2.3.2. RF PCB and transmission-line challenge	6
2.3.3. Frequency measurement	6
2.4. System Integration	7
2.5. Conclusion of the problem analysis	7
3. METHOD	8
3.1. Analysis of the assignment and definition of requirements	8
3.2. Literature study and technical comparison	8
3.3. Engineering calculations and design validation	9
3.4. Schematic design	9
3.5. PCB design	10
3.5.1. Initial RF PCB	10
3.5.2. Final RF PCB redesign	11
3.6. Component selection and ordering	11
3.7. Assembly and testing.	12
3.8. Software development and system integration	12
3.8.1. Approach to software development	13
3.9. Documentation	13
3.10. Conclusion of the methodology	14
4. RESEARCH	14
4.1. System requirements	14
4.1.1. Functional requirements	14
4.1.2. Electrical and RF requirements	15
4.1.3. Safety and reliability requirements	15
4.1.4. Mechanical and integration requirements	16
4.1.5. Power-supply requirements	16
4.1.6. Implementation requirements	16
4.2. Hardware design	17
4.2.1. Overall hardware architecture	17
4.2.2. RF switching matrix	17
4.2.3. RF buses and PCB layout strategy	18
4.2.4. RF connectors	19
4.2.5. Power and SWR measurement	20
4.2.6. Frequency measurement hardware	21
4.2.7. Control board	22
4.2.8. Board separation and shielding	25
4.2.8.1. RF board	25
4.2.8.2. Control board	26
4.2.9. Enclosure and mechanical integration	27

4.2.10. Conclusion of the hardware design	27
4.3. Software design	28
4.3.1. Software design general considerations	28
4.3.2. Relay control	28
4.3.3. State-based switching	29
4.3.4. Safety and interlock logic	29
4.3.5. Power and SWR processing	29
4.3.6. Local user interface	29
4.3.7. Remote communication	30
4.3.8. Watchdog and fault tolerance	30
4.3.9. Conclusion	30
5. RESULTS	31
5.1. Final prototype realization	31
5.1.1. RF-board implementation	31
5.1.2. Control-board implementation	31
5.1.3. Firmware implementation	32
5.2. Bench-level validation	32
5.3. RF measurement results	34
5.4. Power and SWR measurement status	34
5.5. Safety validation	35
5.6. Conclusion of the results chapter	35
6. INTERPRETATION OF RESULTS	35
7. DISCUSSION	36

List of figures

Figure 1. High-power coaxial RF relay	3
Figure 2. Omron G6K-2f-Y Relay	3
Figure 3. Omron G2RL Relay	4
Figure 4. TE Axicom IM06DGR Relay	4

Figure 5. SWR calculation formula.	5
Figure 6 High-level block diagram of the proposed antenna matrix system.	10
Figure 7 selector and fail-safe ground-clamp relay circuit for one antenna channel	17
Figure 8 . AppCAD verification of the final 50 Ω microstrip geometry	18
Figure 9 N-type connectors	19
Figure 10 Bruene coupler schematic	20
Figure 11 Frequency measurement system schematic	21
Figure 12 CYD board	22
Figure 13 CYD interconnection and 1 k Ω I ² C pull-ups.	22
Figure 14 MCP23017 I/O expander	22
Figure 15 TBD62083AFWG relay drivers	23
Figure 16 ADS1115 analog interface	23
Figure 17. TSR-1-2433E 3.3 V step-down regulator	24
Figure 18. TSR-1-2433E 3.3 V step-down regulator	24
Figure 19 TSR 1-2450E 12 V-to-5 V switching regulator.	24
Figure 20 12 V power input connector.	25
Figure 21 RF-board routing, port arrangement and dimensions	26
Figure 22 Control board layout	26
Figure 23 RF board 3D layout	27
Figure 24 Control board 3D layout	27
Figure 25 S11 measurement	34

List of abbreviations and symbols

Abbreviations

ADC — Analog-to-Digital Converter
APPCAD — Application Cad
CAT — Computer Aided Transceiver control
CYD — Cheap Yellow Display
DC — Direct Current
DPDT — Double Pole Double Throw
GPIO — General Purpose Input/Output
HF — High Frequency
I²C — Inter-Integrated Circuit
IC — Integrated Circuit
LCD — Liquid Crystal Display
PCB — Printed Circuit Board
RF — Radio Frequency
RMS — Root Mean Square
SCL — Serial Clock Line
SDA — Serial Data Line
SMD — Surface-Mount Device
SWR — Standing Wave Ratio
TX — Transmitter / Transceiver path
ULN — Darlington transistor driver array
VSWR — Voltage Standing Wave Ratio

Symbols

P — Power
 P_f — Forward power
 P_r — Reflected power
 V_{rms} — Root-mean-square voltage
 I_{rms} — Root-mean-square current
 V_{peak} — Peak voltage
 I_{peak} — Peak current
 Γ — Reflection coefficient
 Z_0 — Characteristic impedance
 ϵ_r — Relative permittivity
 λ — Wavelength
 W — Microstrip width
 H — Substrate height
 T — Copper thickness

Start-up report

In the beginning of this bachelor thesis, the start-up report was completed and accepted by both parties. This document sets out the initial problem statement, the goals of the project, its deliverables, as well as preliminary planning of the project. The initial goal of the thesis was to make an intelligent remote controlled antenna matrix for shortwave radio which is able to switch up to eight antennas to two HF transceivers and provide local and remote control, forward and reflected power measurement, SWR warning, and failsafe behavior.

The start-up report also defines the intended deliverables of the thesis, among which are an RF switch board, a control system, a measurement part, a mechanical assembly, and all the corresponding documentation. Apart from it, an initial phased planning was made, which included phases such as analysis, system design, hardware design, firmware programming, prototype making, testing, integration, and writing the thesis.

However, in the process of developing the thesis, several issues concerning technical decisions required further analysis. These included relay selection, RF circuitry, grounding technique, and measurement approach. Therefore, the realization of the project was evolving compared to what was planned initially but it remained consistent with the primary tasks set out in the start-up report.

1. Introduction

Modern HF radio stations may contain more than one antenna in order to enhance the communication performance in different frequency bands and with different propagation conditions. Selection of an appropriate antenna for each case is needed for efficient operation, minimum reflected power and overall performance improvement. However, switching between multiple antennas in HF stations is often manual.

The aim of this bachelor thesis is to design and implement a remotely controlled intelligent antenna matrix for shortwave radio. The device is meant to switch eight antennas between two HF transceivers while guaranteeing safe operation at the RF levels. Besides switching itself, the device shall measure voltage standing wave ratio (VSWR), antenna power and optional frequency. Moreover, local status monitoring and remote-control functionality are desirable.

One of the biggest challenges in this project lies in combining all those functions together while considering the safety measures for such high-power RF devices. Switching circuits must reliably switch between antennas but should not cause unsafe relay states, wrong connections, improper antenna selection and switching when the transmitter is transmitting. Intended for remote control and monitoring usage, it should also provide watchdog and default state mechanisms as well as safe grounding of unconnected antennas.

Besides that, another critical issue of the project lies in RF signal handling and proper design of embedded circuitry to minimize electromagnetic interference and other effects caused by high RF level operation. It will be required to make proper decisions on separation between RF signal path components, embedded controllers and PCB layout in order to achieve desired results.

The above-mentioned tasks are typical for such a project, making it multi-disciplinary work. They include RF circuit design, measurement electronics, PCB design and embedded software design among others. Therefore, the main objective of this work will be development of a practical device for remote controlling HF radio antennas with additional monitoring capabilities.

This report provides a comprehensive overview of the entire design process from the initial idea and system requirements analysis to the design of PCB layout and assembly of the actual prototype with its evaluation and discussion.

2. Problem statement

The major aim of this project is to design an intelligent antenna matrix that would allow connecting eight HF antennas to two HF transceivers in a safe way. At an initial overview, this task looks like a pure problem of switching. However, it actually comprises several aspects related to high-frequency systems engineering: high power levels, reliable remote switching, low switching losses, and protection features, among others.

2.1. Operational need

Multiple antennas are used at HF radio stations since no single antenna is always good for all kinds of operations. The choice of the antenna is determined by the frequency range, propagation conditions, mode of operation, and the required radiation pattern. Switching of antennas is traditionally done by a human operator, and while it seems simple for one antenna and one radio transceiver, it becomes problematic for multiple antennas and two radios. For remote stations, the possibility to switch antennas manually becomes impossible, and, therefore, there is a necessity to design an automatically controllable antenna matrix.

Hence, the major requirement for the system is its ability to implement flexible antenna routing in a safe manner.

2.2. Main technical difficulty

Probably the most challenging part of the proposed system development is the high-frequency switching stage. According to the specifications, the solution shall support HF transceivers capable of transmitting power up to 1 kW. Such high power implies the use of specially designed high-frequency relays with contact capabilities both in terms of current capacity and reliability.

For a 50 Ω system at 1 kW, the matched RMS voltage and current are:

$$V_{rms} = \sqrt{P \cdot R} = \sqrt{1000 \cdot 50} \approx 223.6 \text{ V} \quad I_{rms} = \sqrt{\frac{P}{R}} = \sqrt{\frac{1000}{50}} \approx 4.47 \text{ A}$$

This already points out that the RF path must carry several amperes continuously.

However, HF systems are not always perfectly matched. An important design consideration is the case of VSWR = 3:1.

For a VSWR of 3:1, the reflection coefficient magnitude is:

$$|\Gamma| = \frac{VSWR - 1}{VSWR + 1} = \frac{3 - 1}{3 + 1} = 0.5$$

The maximum standing-wave voltage and current are then multiplied by:

$$1 + |\Gamma| = 1.5$$

This leads to:

$$V_{max,rms} \approx 223.6 \times 1.5 \approx 335.4 \text{ V} \quad I_{max,rms} \approx 4.47 \times 1.5 \approx 6.71 \text{ A}$$

The corresponding peak values are:

$$V_{max,peak} \approx 335.4 \times \sqrt{2} \approx 474 \text{ V} \quad I_{max,peak} \approx 6.71 \times \sqrt{2} \approx 9.48 \text{ A}$$

These calculations show that the relay contacts and PCB traces may experience voltages and currents of approximately 335 V RMS, 474 V peak, 6.7 A RMS, and 9.5 A peak under mismatch conditions.

This makes relay selection one of the most difficult parts of the design.

2.2.1. Relay selection problem

In an ideal case, the technical solution should involve the usage of high-power coaxial RF relays like **Figure 1**.

These relays are purposefully built for RF switching, with controlled impedance, better voltage spacing and power handling. Unfortunately, such relays are prohibitively expensive and would dominate the cost of the whole project. Thus, there was a need for a compromise between technical capabilities, availability, size and price.

As a matter of fact, during the course of the design, the relay selection problem occurred multiple times. This became one of the central challenges of the project.



Figure 1. High-power coaxial RF relay

A. Initial attempt: Omron G6K relay

Initially, the optimal solution seemed to be the usage of the Omron G6K relay seen in **Figure 2**. It is one of the smallest relays available, and hence it is attractive from a layout perspective. Using G6K, it would be possible to create a compact board with 8×2 switching matrix.

However, after further consideration of this relay, it turned out that it belongs to small-signal relays and thus is not suitable for high power applications. Its current and voltage ratings, as well as its contact spacing and mechanical robustness were insufficient for the purposes of this project. Despite the usage of both poles in parallel, the relay remained insufficiently powerful.

Thus, a very compact relay seems to be very attractive, but it may not be enough to build a reliable 1 kW HF RF switch.



Figure 2. Omron G6K-2f-Y Relay

B. Next stage: Search for an alternative

After excluding the G6K as the relay for the RF circuit, the next task to address became whether there exists a larger but still relatively cheap relay with significantly better current ratings and voltage tolerance.

At that stage, one of the main comments from the supervisor was that each contact must be capable of carrying at least 3 A current. In particular, this was due to the fact that the design uses DPDT relays with both poles working in parallel. Having at least 3 A per pole makes the current sharing more realistic.

i. Larger relay: Omron G2RL

One of the next relays to consider was the Omron G2RL family (**Figure 3**).

It was better suited in terms of current rating and mechanical robustness compared to the G6K relay. It appears to be a true power relay and provides much greater confidence in regard to its ability to function under high power.

Unfortunately, it also has one serious disadvantage – size. It is significantly larger than G6K relay and thus makes the layout of 8 channels much more complicated. This creates additional pressure on PCB size and routing.



Figure 3. Omron G2RL Relay

ii. SMD Relay alternative: IM06DGR

As the need for compactness of the board did not disappear and an SMD solution looked promising, there was a need to find a proper relay of the same family. One of the options was the family of TE Axicom IM series relays. In particular, the IM06DGR model (**Figure 4**) is a DPDT, 12V and 5A SMD relay.

This relay is a good replacement for the G6K as it has much more power capacity compared to the initial design solution. At the same time, it is smaller than G2RL and thus offers advantages in terms of size.

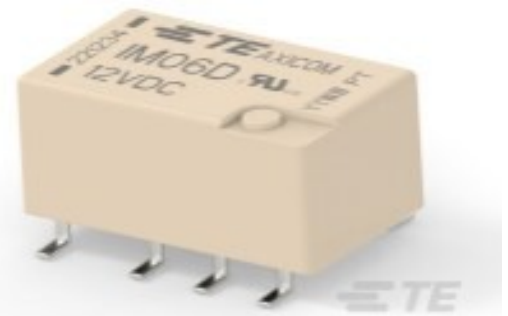


Figure 4. TE Axicom IM06DGR Relay

However, similar to other relays of this family, IM06DGR is still not optimized for RF operation. Thus, it cannot be installed arbitrarily and requires rules such as:

- Both poles in parallel
- No hot switching
- Short RF paths
- Good 50Ω bus layout
- Software interlocks to prevent entering an invalid state.

Thus, regarding relay selection there appeared a clear need for trade-offs between RF capabilities, current ratings, package size, mechanical strength, PCB space and price.

2.2.2. Ground-clamp relay problem

Apart from the mentioned problems, another challenge in the relay matrix design involved implementing the ground-clamp relays for the unused antennas. The function of the ground clamp is to make sure that any antenna not being used is electrically grounded, since leaving antennas floating creates potential coupling issues and adds unpredictability and unreliability to the overall system behavior.

At the beginning of the design process, it was realized that there was a mistake in the way the wiring of the clamp relay is organized. Namely, the ground has been placed to the wrong contact side, which means that

the antenna will remain grounded in its default position. As a result, special attention was paid to understanding how the COM, NC, and NO contacts in a typical relay work. The right solution was found: the antenna node must be wired to the common contacts, while the ground – to the normally-open contacts, so the antenna is grounded only in the case when the clamp relay is closed on purpose.

- **Use of the same relay type for ground clamp**

One of the other decisions during the design process was to choose the same relay family for both the switching relays and the ground clamp relays. In particular, at first sight it looked appealing to use a smaller relay as the clamp element. The reasoning behind such a decision was the idea of saving PCB area. However, since the clamp relay is directly connected to the antenna RF node as well, the clamp relay may potentially be subjected to RF-related voltage and current stress. Using a weak relay as the clamp relay would create an additional weak point in the RF chain, and, therefore, a stronger relay should be used instead.

2.3. Measurements and integration difficulties

As stated above, the project includes several additional functions beyond the antenna switching matrix itself. The final system integrates forward and reflected signal measurement, SWR calculation, safety and interlocking logic, local control and remote control.

Frequency measurement was also investigated during development as an optional additional function. However, practical testing showed that the implemented solution was not sufficiently reliable across the required HF operating range, and it was therefore removed from the final implementation.

2.3.1. Forward and reflected power measurements and SWR calculation

An important requirement of the RF system is the ability to determine the forward and reflected components of the transceiver output in order to detect mismatch conditions.

The Bruene directional coupler produces analog detector voltages corresponding to the forward and reflected RF components. These voltages are digitized using the ADS1115 and subsequently processed by the controller. After calibration, they can be used to estimate transmitted and reflected power and to calculate the Standing Wave Ratio (SWR).

When the reflected component is small compared with the forward component, the antenna system is well matched and the SWR approaches 1:1. As the reflected component increases, the SWR increases, indicating a poorer impedance match between the transmitter and antenna system.

$$SWR = \frac{1 + \sqrt{P_r / P_f}}{1 - \sqrt{P_r / P_f}}$$

Figure 5. SWR calculation formula.

where P_f is the forward power and P_r is the reflected power. This relation is based on the reflection coefficient, which can be expressed in terms of the ratio between reflected and forward power. When the reflected power is very small compared to the forward power, the antenna system is well matched and the

SWR approaches 1:1. As the reflected power increases, the SWR also increases, indicating a poorer impedance match between the transmitter and the antenna.

This formula is particularly useful in the proposed system because the forward and reflected outputs of the directional coupler can be measured separately. Once these two quantities are available, the controller can estimate the SWR in software and use it for monitoring, warning, or protection purposes.

For this purpose, two possible directional coupler topologies were considered: the Bruene coupler and the tandem coupler. Although the tandem coupler is known for its good directivity, it is more complex to design and manufacture. The Bruene coupler was considered a more practical solution for the present prototype because it combines a voltage sample and a current sample to separate the forward and reflected components of the RF signal. In addition, it is a well-established approach in HF wattmeter and SWR meter designs.

For these reasons, the Bruene coupler was selected for this project. It provides the required measurement principle while being simpler to implement and requiring fewer components than the tandem coupler. Furthermore, the use of adjustable trimmer capacitors makes it possible to balance the bridge during testing, which improves directivity and increases the accuracy of the reflected-power measurement.

2.3.2. RF PCB and transmission-line challenge

Another major challenge is the design of the PCB transmission paths themselves. The antenna matrix forms part of a 50 Ω RF system, meaning that the PCB traces, relay transitions and connector interfaces must maintain this impedance as closely as possible.

The first prototype used a 4-layer PCB with an internal ground plane located close to the top RF layer. Although this provided a well-defined ground reference, the resulting 50 Ω microstrip traces were relatively narrow, approximately 0.77 mm after correction of the PCB stack-up calculation. This geometry was not considered ideal for continuous high-power operation.

Practical VNA measurements of the first prototype at approximately 14.5 MHz resulted in a return loss of approximately 10.63 dB and an insertion loss of approximately 0.46 dB. Isolation and crosstalk were approximately 28.43 dB and 30.46 dB respectively.

The measurements demonstrated that the switching matrix was functional but also showed that the RF transmission path required further optimization. For this reason, the RF board was redesigned as a 2-layer PCB with a continuous bottom ground plane and significantly wider 50 Ω RF conductors.

This illustrates that the RF performance of the antenna matrix depends not only on relay selection, but also on PCB stack-up, conductor geometry, grounding, connector transitions and parasitic effects.

2.3.3. Frequency measurement

One more desirable functionality of the system is the ability to measure the transmitted RF frequency. At the initial design stage, CAT control was proposed as one of the alternatives because it allowed reading the operating frequency from the transceiver directly. However, this option required the presence of a proper communication interface and particular protocol of operation of the radio device. Therefore, this method could hardly be considered universal enough for a prototype designed to be as independent from the exact transceiver model as possible.

For this reason, it was decided to follow the hardware implementation of the RF frequency measurement. This functionality is provided by the combination of an RF pickup capacitor, an attenuation and protection stage, and the LMV7219 comparator. The RF pickup capacitor takes the sample of the RF signal sent from

the transceiver, and then this signal is attenuated and protected in order to supply it to the comparator, where it is transformed into a pulse stream of digital values readable by the controller.

On the one hand, the principle behind the measurement process seems rather simple, but, on the other hand, implementing this functionality is somewhat problematic. In particular, due to the limited availability of free GPIO pins of the selected CYD controller platform, there is little choice when it comes to implementing the frequency measurement functionality with two inputs.

2.4. System Integration

The matrix cannot be treated as a stand-alone RF switch, because it must also integrate measurement, control, remote communication, and safety logic. As a result, the project must be considered as a complete RF and embedded control system rather than only a relay-based switching network.

2.5. Conclusion of the problem analysis

Therefore, the problem tackled in this thesis does not just relate to switching antennas; it relates to the ability to do so safely, efficiently, and economically while under the high-power conditions characteristic of HF. One of the main challenges encountered during the course of the project concerned the selection of the relay itself, due to the often-competing factors of power capacity, size, cost, manufacturing ability, and PCB space.

Moreover, there was the added challenge of integrating RF switching with power measurement, SWR measurement, frequency measurement, embedded processing, and remote control. Each of these systems presented additional design considerations and, when combined, added greatly to the complexity of the problem to solve a simple RF switch.

Therefore, the numerous design changes that took place throughout the duration of this project were not necessarily an indicator of poor planning, but instead resulted from increasingly accurate calculations, guidance by the thesis advisor, and a greater understanding of the RF and system limitations.

3. Method

The Smart Antenna system was developed using an iterative engineering methodology comprising requirements analysis, literature review, engineering calculations, schematic and PCB design, component selection, staged assembly, firmware development, and verification. Findings obtained during calculation, assembly, and measurement were incorporated into subsequent design revisions. Consequently, the final prototype resulted from successive cycles of design, implementation, testing, and improvement rather than from a strictly linear development process.

3.1. Analysis of the assignment and definition of requirements

The assignment was first translated into functional and technical requirements. The principal functional requirement was to route any of eight HF antennas to either of two transceivers while preventing an antenna from being connected to both RF buses simultaneously. Additional requirements included local touchscreen operation, remote control through Blynk, acquisition of forward and reflected detector signals, calculation of power and SWR, an optional frequency-detection function, defined fail-safe states, manual transmitter locks, and watchdog recovery.

The technical requirements included nominal 50 Ω RF paths, a high-power design target of up to 1 kW, reliable relay operation, safe behavior under mismatched-load conditions, suitable RF connectors, and adequate separation between the RF and digital-control sections. Mechanical constraints relating to the PCB and enclosure were also considered.

These requirements served as design targets. Compliance with the final RF-performance and high-power requirements could only be established through measurements and was therefore evaluated separately from the initial design process.

3.2. Literature study and technical comparison

A literature review and technical comparison were conducted to identify suitable solutions for the critical subsystems. The investigation considered:

- relay types and contact configurations for the RF switching matrix
- grounding arrangements for unused antennas
- directional-coupler topologies for forward- and reflected-power detection
- possible frequency-detection methods

The Bruene and tandem-match coupler topologies were evaluated for the power-measurement subsystem. The Bruene coupler was selected because it offered a compact implementation suitable for PCB integration and allowed practical adjustment using trimmer capacitors.

CAT communication with the transceivers was initially considered for frequency acquisition. However, this approach would have made the system dependent on the communication protocol of each connected transceiver. A hardware-based alternative was therefore investigated, consisting of an RF pickup, attenuation and protection components, and a comparator for generating a logic-level pulse signal. This circuit remained an optional hardware provision rather than a validated function of the final system.

3.3. Engineering calculations and design validation

Engineering calculations were performed before final component selection and PCB layout. RF voltage and current levels were determined for a $50\ \Omega$ system at a maximum design power of 1 kW. An additional calculation was performed for a VSWR of 3:1 to estimate the increased electrical stress that could occur under mismatched-load conditions.

The results were used to assess relay contact requirements, conductor dimensions, connector suitability, grounding arrangements, component voltage ratings, and appropriate design margins. These calculations supported component selection but did not, by themselves, demonstrate operation at the full design power.

Microstrip calculations were also performed using AppCAD to determine the conductor dimensions required for an approximately $50\ \Omega$ characteristic impedance. The calculated geometries were subsequently compared with Vector Network Analyzer measurements of the first assembled RF board. This comparison was an important part of the iterative design process.

3.4. Schematic design

Before producing complete schematics, the system was represented using a functional block diagram.

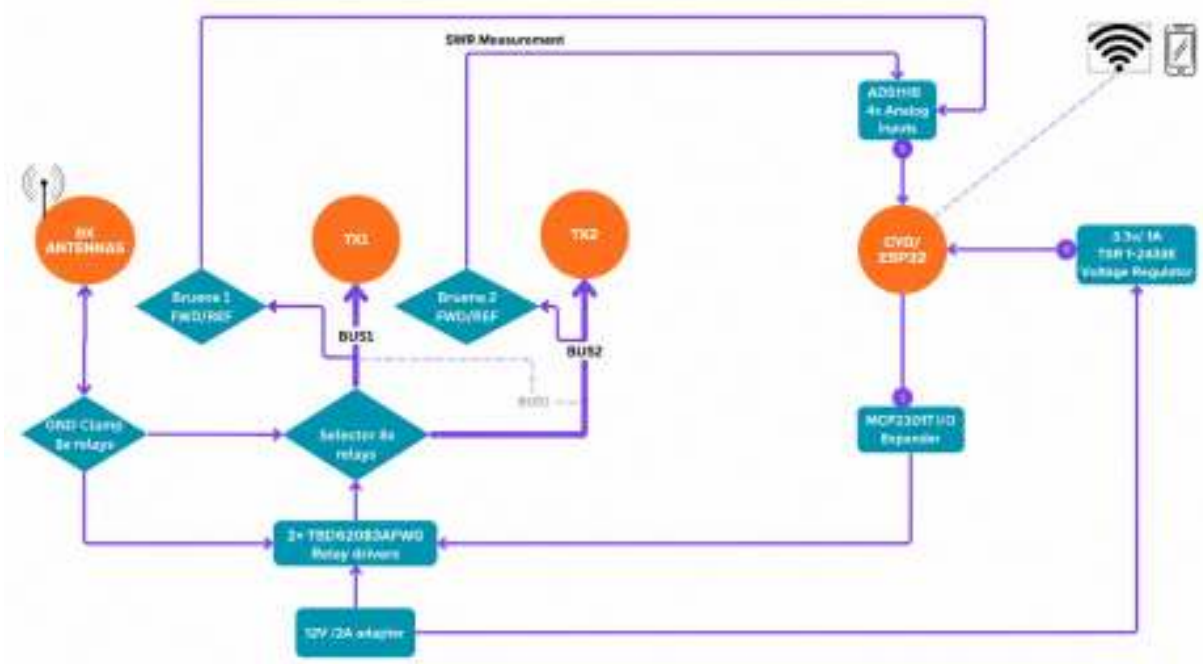


Figure 6 High-level block diagram of the proposed antenna matrix system.

The above block diagram explains how the signal flow and interaction among the different systems occurs. Each antenna connection passes through a ground-clamp relay and a selector relay. The selector relay connects the antenna to BUS1 or BUS2, while the ground-clamp relay connects the antenna to ground when it is not assigned to either bus. In the fail-safe, de-energized state, the normally closed contacts of the ground-clamp relay park the antenna at ground.

Each RF bus passes through a Bruene directional coupler before reaching its respective transceiver connector. The couplers generate analog signals proportional to the forward and reflected RF components. These signals are routed to the control board for acquisition by the ADS1115. The optional frequency-detection circuits use an RF pickup and comparator to produce logic-level signals, although frequency counting is not implemented in the final firmware.

The control board contains the ESP32-based CYD interface, MCP23017 I²C port expander, ADS1115 analog-to-digital converter, TBD62083AFWG relay-driver arrays, power converters, and the associated interface circuitry.

3.5. PCB design

The system was divided into two PCBs: an RF board and a control board. This separation simplified the layout and reduced interaction between the high-power RF paths, relay-coil signals, analog detector outputs, and low-voltage digital electronics.

3.5.1. Initial RF PCB

The first RF prototype was implemented as a four-layer PCB. The main RF paths were routed on the top layer, with an internal layer immediately below them serving as the RF ground reference. The remaining layers were used for power distribution and control routing.

After the stack-up parameters were corrected, the calculated width required for an approximately 50 Ω microstrip was found to be approximately 0.77 mm because of the short distance between the top layer and the internal ground plane. Although this geometry could theoretically provide the required characteristic

impedance, the narrow conductors increased resistance and current density and were not considered appropriate for the intended high-power design target.

The assembled board was measured using a Vector Network Analyzer. At approximately 14.5 MHz, the following results were obtained:

- S11 approximately -10.63 dB, corresponding to a return loss of 10.63 dB and a VSWR of approximately 1.83.
- S21 approximately -0.46 dB, corresponding to an insertion loss of 0.46 dB.
- Isolated-path transmission: approximately -28.43 dB.
- Adjacent-path coupling: approximately -30.46 dB.

These measurements demonstrated that the switching principle was functional. However, the board did not meet the project targets of a VSWR below 1.1 and an insertion loss below 0.2 dB. The measurements therefore provided the basis for redesigning the RF board.

3.5.2. Final RF PCB redesign

The final RF board was redesigned as a two-layer PCB using approximately 1.6 mm FR-4, a 1.509 mm dielectric thickness, a relative permittivity of 4.6, and 0.03556 mm copper thickness. An AppCAD calculation produced a characteristic impedance of approximately $50.52\ \Omega$ at 30 MHz for a microstrip width of 2.7 mm.

The redesigned layout uses the top layer for the RF paths and relay connections and a continuous bottom-layer ground plane as the RF reference. The RF routes were shortened where possible, connector grounding was improved, and avoidable discontinuities were reduced. Compared with the initial board, the wider conductors also reduce conductor resistance and current density. These improvements support the high-power design objective but do not independently prove 1 kW capability.

The N-type connectors are mechanically attached to the metal enclosure cover, while the PCB is positioned close to the cover using metal standoffs. This arrangement provides short connections between the connector bodies, PCB ground, and metal enclosure.

To satisfy the original VSWR target of less than 1.1, the corresponding return-loss target is greater than approximately 26.4 dB. The insertion-loss target remains below 0.2 dB. At the time of writing.

3.6. Component selection and ordering

After schematic development and preliminary PCB design, components were selected and ordered.

Component selection was based on several factors:

- Electrical ratings
- RF suitability
- Package dimensions
- Availability
- PCB area
- Cost
- Ease of assembly

Several components changed during development as better information became available.

For example, the relay selection evolved from smaller signal relays toward the stronger IM06DGR relay family.

The relay-driver stage also changed from the ULN2803 Darlington array to the TBD62083A DMOS array because of component availability and its lower-loss output structure.

Trimmer capacitors and ferrite-core components used in the Bruene coupler were also revised as part of the RF-board redesign.

3.7. Assembly and testing.

The prototype was assembled in stages so that individual subsystems could be examined before complete integration. Initial tests covered the 12 V, 5 V, and 3.3 V supply rails, followed by I²C communication, port-expander outputs, relay-driver operation, relay-state control, analog acquisition, touchscreen interaction, Blynk communication, and watchdog behavior.

During integration, the I²C bus proved unreliable with the initially selected 4.7 k Ω pull-up resistors. The final configuration uses GPIO27 as SDA, GPIO22 as SCL, and 1 k Ω pull-up resistors to 3.3 V. This modification improved the signal rise time on the loaded bus and allowed the MCP23017 at address 0x20 and ADS1115 at address 0x48 to communicate reliably.

Bench-level testing confirmed the following ADS1115 channel allocation:

- AIN0: TX1 forward-detector signal;
- AIN1: TX1 reflected-detector signal;
- AIN2: TX2 forward-detector signal; and
- AIN3: TX2 reflected-detector signal.

Small residual or unstable readings were observed when the detector outputs were inactive. Four 100 k Ω pull-down resistors were therefore added to provide a defined zero-voltage state and discharge path without significantly loading the detector outputs.

The control tests demonstrated valid antenna-state management, one occupant per RF bus, manual transmitter locks, boot-time parking, local and remote command handling, and basic watchdog recovery. These were bench-level functional tests and did not demonstrate automatic RF or PTT interlocking, contact-position feedback, or complete hot-switch prevention under transmitted power.

The first RF PCB was characterized using a VNA, and the results directly informed the transition from the four-layer layout to the final two-layer design.

3.8. Software development and system integration

The software included:

- Relay control,
- Antenna-state management,
- BUS1/BUS2 interlocking,
- TX lock functionality,
- ADS1115 measurement processing,
- Power estimation,
- SWR calculation,
- Touchscreen interaction,
- Serial diagnostics,
- Blynk remote control.

The CYD platform was used as the main controller and local user interface.

The limited number of available GPIO pins introduced integration constraints because several pins were already required by the display and touchscreen.

Software development was performed together with hardware testing so that each hardware subsystem could be verified independently before complete system integration.

The initial firmware also included experimental frequency measurement using ESP32 pulse counting. This functionality was removed from the final software when the frequency-measurement hardware was removed.

3.8.1. Approach to software development

Software development was performed incrementally.

The first functions implemented were:

- I²C communication,
- MCP23017 detection,
- Relay output control,
- ADS1115 communication,
- Individual ADC channel reading.

After these basic functions were verified, additional functionality was added:

- Antenna selection,
- BUS1/BUS2 routing,
- Ground-clamp control,
- Switching interlocks,
- TX lock logic,
- Power calculation,
- SWR calculation,
- Touchscreen control,
- Remote Blynk control.
- Watchdog.

Serial diagnostic commands were also used during development to test relay states and analog measurements independently from the graphical interface.

This incremental approach made it easier to isolate hardware and software problems during integration.

3.9. Documentation

The thesis documentation was developed in parallel with the hardware and software implementation.

The documentation includes:

- Design decisions,
- Engineering calculations,
- Component-selection changes,
- Schematic revisions,
- PCB revisions,
- Software development,
- Measurement results,
- Problems encountered during testing,
- Improvements made as a result of those tests.

This approach ensures that the thesis describes not only the final implementation but also the engineering process that led to it.

3.10. Conclusion of the methodology

The methodology used in this thesis can be described as an iterative engineering process.

The project did not follow a completely linear development path. Several design choices were modified after calculations and practical testing revealed limitations in the initial implementation.

Examples include:

- Changes in relay selection,
- Replacement of the ULN2803 with the TBD62083A,
- Addition of 100 k Ω ADC pull-down resistors,
- Removal of the unreliable frequency-measurement subsystem,
- Redesign of the RF PCB from a 4-layer to a 2-layer structure following VNA measurements.

These changes were based on measurements, calculations, component evaluation and practical testing.

The final design therefore resulted from repeated cycles of design, implementation, measurement and improvement rather than from a single fixed design process.

4. Research

4.1. System requirements

The system requirements were derived from the project description, the intended application of the antenna matrix, and the technical limitations identified during the analysis phase.

The requirements define the intended operational and technical capabilities of the prototype and are divided into the following categories:

1. Functional requirements,
2. Electrical and RF requirements,
3. Safety and reliability requirements,
4. Mechanical and integration requirements,
5. Power-supply requirements,
6. Implementation requirements.

Some requirements represent final implemented functionality, while others represent design targets that require further validation, particularly those related to continuous operation at 1 kW RF power.

4.1.1. Functional requirements

The system shall provide the capability to route multiple HF antennas between two HF transceivers.

The functional requirements are:

- The system shall support eight antenna inputs.
- The system shall support two HF transceiver paths.
- Each antenna shall be selectable for either BUS1 or BUS2.
- An antenna that is not assigned to a transceiver shall be placed in a controlled safe state using the ground-clamp function.

- The system shall provide local monitoring and control using the CYD touchscreen.
- The system shall provide remote control through the Blynk interface.
- The system shall measure separate forward and reflected detector signals for both transceiver paths.
- The controller shall use these measurements to estimate transmitted power and calculate SWR.
- The system shall include interlocking logic to prevent conflicting or unsafe antenna-routing states.
- The system shall provide TX-lock functionality to prevent antenna switching while a transceiver path is considered active.

Frequency measurement was initially included as an optional design objective. A hardware-based implementation using an RF pickup and comparator was developed and tested, but it did not provide sufficiently reliable operation across the intended HF range. It was therefore removed from the final implementation.

4.1.2. Electrical and RF requirements

Because the system is intended for HF transmission at relatively high RF power, the electrical and RF requirements are particularly important.

The main requirements are:

- The RF system shall be designed for a nominal impedance of 50 Ω .
- The RF switching hardware shall be designed with a target power-handling capability of up to 1 kW continuous RF power.
- The RF paths shall be designed to minimize insertion loss.
- The RF layout shall minimize impedance discontinuities, reflections and unnecessary parasitic effects.
- The main RF buses shall use short and direct transmission paths.
- Approximately 50 Ω microstrip geometry shall be used for the principal RF traces.
- The redesigned RF board shall use wider RF conductors suitable for high-current operation.
- The bottom layer of the redesigned RF PCB shall provide a continuous ground reference underneath the principal RF paths.
- RF circuits shall be arranged to minimize coupling between BUS1 and BUS2 and between the RF section and the digital control circuitry.
- The directional couplers shall provide separate forward and reflected detector signals for both transceiver paths.
- Relay contacts, PCB conductors, connectors and measurement circuitry shall be selected with sufficient voltage and current margins for expected mismatch conditions.

The 1 kW value represents the design target of the system. Final continuous operation at this power level requires high-power validation of the redesigned RF board.

4.1.3. Safety and reliability requirements

Remote control and high RF power require additional measures to prevent unsafe switching conditions.

The following safety and reliability requirements were defined:

- The system shall prevent invalid relay combinations through software interlocking.
- Only one antenna shall be assigned to each RF bus at a time.
- An antenna shall not be simultaneously connected to both transceiver paths.
- TX-lock functionality shall be provided to prevent switching on a bus while that transceiver path is considered active.
- Unused antennas shall be placed in a defined and controlled ground-clamped state.

- The system shall monitor SWR so that high-mismatch conditions can be identified by the operator.
- The design shall minimize the possibility of unsafe switching states resulting from operator error.
- Relay switching shall follow a controlled sequence so that antennas are disconnected or parked before being transferred between RF buses.

The original project requirements also considered automatic fail-safe behavior during loss of power and watchdog functionality for remote operation. These functions should only be considered part of the final implementation if they are fully implemented and verified in the final hardware and software.

4.1.4. Mechanical and integration requirements

The mechanical design must support both the high-power RF section and the low-voltage control electronics.

The requirements include:

- The system shall use separate RF and control PCBs.
- The RF and digital control sections shall be physically separated in order to reduce unwanted RF coupling.
- The system shall be installed in a conductive metal enclosure suitable for RF shielding.
- The enclosure shall accommodate eight antenna connectors and two transceiver connectors.
- The N-connectors shall be mechanically supported by the metal enclosure rather than only by the PCB.
- The redesigned RF PCB shall be mounted close to the metal enclosure cover using metal standoffs.
- The grounding arrangement shall provide short and low-impedance connections between the PCB ground, connector bodies and enclosure.
- The mechanical layout shall allow access for assembly, testing, tuning and debugging.

4.1.5. Power-supply requirements

The system contains both 12 V relay circuitry and low-voltage digital electronics.

The power-supply requirements are:

- The system shall operate from an external 12 V supply.
- The 12 V rail shall provide the required current for the relay coils.
- The digital control and measurement electronics shall be supplied from the required regulated low-voltage rails.
- Relay-current return paths shall be arranged so that switching currents do not significantly disturb the digital or analog measurement circuitry.
- Appropriate decoupling shall be provided close to the digital ICs and measurement devices.
- The power distribution shall maintain a common and controlled ground reference between the control circuitry and the RF measurement interface.

4.1.6. Implementation requirements

In addition to the electrical and functional requirements, the prototype shall remain practical to manufacture and test.

The implementation requirements are:

- The design shall remain economically realistic for a prototype.
- Component selection shall balance electrical performance, RF suitability, availability, size and cost.

- Components that are difficult to source or unsuitable for the final design may be replaced by technically appropriate alternatives.
- The PCB shall be manufacturable using commonly available PCB fabrication technology.
- The design shall allow manual assembly, inspection and repair.
- The prototype shall be testable in stages so that relay control, ADC acquisition, RF measurements, communication and software functions can be verified independently.
- The design shall support iterative modification when practical measurements reveal limitations in the initial implementation.

The final prototype therefore represents an engineering compromise between high-power RF performance, cost, physical size, component availability and practical manufacturability.

4.2. Hardware design

The hardware of the proposed antenna matrix was designed to combine high-power HF switching, RF measurement, digital control, and safe system operation in a single prototype. Due to the combination of RF and digital parts, the hardware was divided into two boards – the RF board and the control board. Such separation reduces interference between RF and digital parts and meets the project requirement according to which the RF part should be well-shielded from the microcontroller part.

4.2.1. Overall hardware architecture

The complete hardware system consists of the following building blocks:

- Antenna switching matrix relays,
- Ground-clamp relays for unused antennas,
- Directional couplers for forward and reflected power measurement,
- Frequency pickup and conditioning circuits,
- Relay driver circuitry,
- Analog measurement interface,
- CYD-based control platform,
- Power supply and regulation.

The RF board contains the antenna switching matrix, two Bruene couplers, and the optional frequency pickup circuits. The control board contains the CYD controller, MCP23017 I/O expansion, TBD62083AFWG relay drivers, ADS1115 analog acquisition, inter-board connectors, and the 12 V-to-5 V and 12 V-to-3.3 V supplies.

4.2.2. RF switching matrix

The main function of the hardware is a switching outputs. Each antenna path contains the following:

- A. Selector relay – to connect the antenna to
- B. Ground-clamp relay – to ground the antenna

Both relay functions are illustrated in **Figure 7**.

This configuration allows each antenna to be placed

- Connected to transceiver Bus 1,
- Connected to transceiver Bus 2,
- Grounded in the parked state.

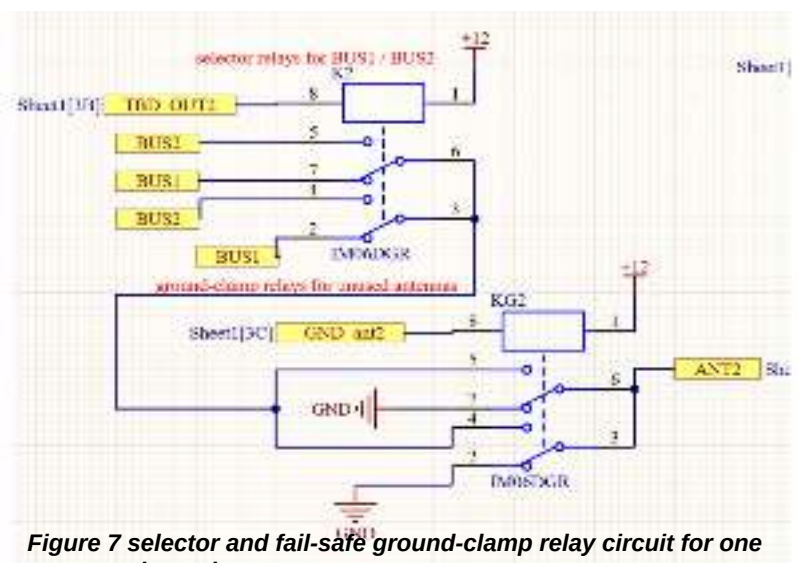


Figure 7 selector and fail-safe ground-clamp relay circuit for one antenna channel

A. Selector relays

The RF switching matrix uses 12 V IM06DGR DPDT relays. The two poles of every selector relay are wired in parallel: corresponding BUS1 throws are paired, corresponding BUS2 throws are paired, and the two common contacts form the antenna-side path. This reduces effective contact resistance and shares RF current between contacts. The selector function chooses BUS1 or BUS2; it does not provide the grounded state by itself.

B. Ground-clamp relays

As unused antennas should not be left floating, each antenna is equipped with a ground-clamp relay. Its purpose is to reliably ground the unused antenna.

Clamp relays are also implemented with both poles in parallel, since they are directly connected to the antenna RF node and thus belong to the RF power path. Though it seems optimal to use a smaller relay for the clamp function, such implementation would introduce a weak link into the switching network. Therefore, the same relay family was selected for both functions.

4.2.3. RF buses and PCB layout strategy

The RF board was designed around two main RF buses:

- BUS1
- BUS2

These buses represent the two transceiver paths through the switching matrix. In order to improve the RF behavior of the design, the layout was developed around short microstrip traces with a solid ground reference underneath. This approach was selected to obtain a more controlled impedance and to reduce unwanted reflections and parasitic effects in the RF path.

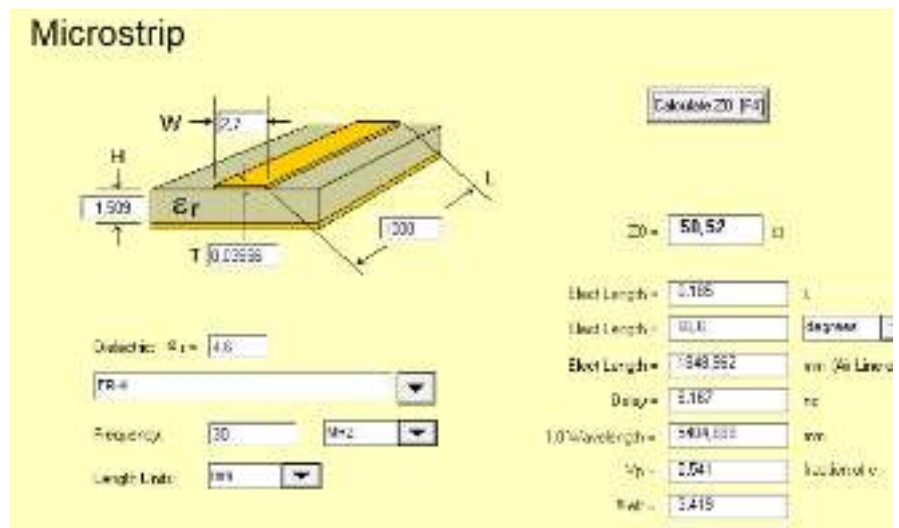


Figure 8 . AppCAD verification of the final 50 Ω microstrip geometry

The final RF-bus geometry was recalculated in AppCAD using a 1.509 mm FR-4 dielectric, relative permittivity 4.6, 0.03556 mm copper, and a 30 MHz design frequency. A 2.7 mm top-layer trace gives a calculated characteristic impedance of 50.52 Ω and an effective relative permittivity of approximately 3.419. The trace is referenced to the continuous bottom ground plane. The layout also uses short relay stubs, approximately 0.10 mm minimum clearance where fabrication permits, and ground-via stitching around the RF region.

Specific attention was given to:

- Relay orientation,
- Short RF trace lengths,
- Sufficiently wide copper traces,
- Minimization of parasitic bus length,
- Strong grounding.

The RF traces were designed to be as straight and short as possible between the antenna nodes, relay contacts, buses, and transceiver outputs. In addition, the ground-clamp paths were also kept short and low-impedance in order to ensure an effective grounding function for unused antennas.

4.2.4. RF connectors

For the connections to the external RF circuits, the chosen connectors are N-type connectors for the antenna connections as well as for the transceiver connections. This choice was made because the system is intended to operate at relatively high power in the HF range, where mechanical robustness and reliable RF connection are essential.

The N-type connectors seem to be preferable over the SMB connectors for several reasons. First, they offer significantly improved mechanical strength. Secondly, they are better designed for use at high RF power, whereas SMB connectors are typically used in signal-level or compact RF systems.

Thirdly, the N-type connectors are frequently used in practice and are more suitable for RF systems installed in a shielded metal enclosure.



Figure 9 N-type connectors

4.2.5. Power and SWR measurement

To monitor the RF performance of the system, two directional measurement circuits are included, one for each transceiver path. These circuits provide information about:

- Forward power,
- Reflected power,
- The standing wave ratio (SWR).

The selected measurement principle is based on the Bruene coupler. This coupler was chosen because it provides a practical and relatively simple solution for separating the forward and reflected components of the RF signal in an HF system. It is therefore well suited to the requirements of this prototype.

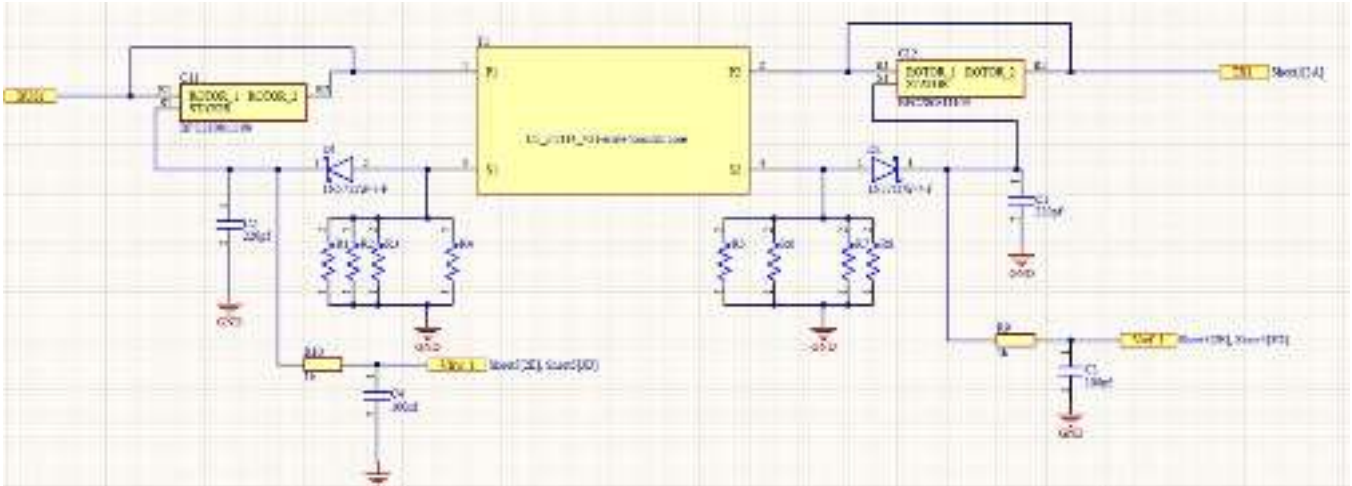


Figure 10 Bruene coupler schematic

The Bruene coupler shown in *Figure 10*. consists of:

- A current transformer built around a ferrite toroidal core,
- A voltage sampling network,
- Bridge balancing components,
- Rectifying diodes,
- Filtering circuitry.

The bridge sampling capacitors are adjustable 1–3.5 pF trimmers so that each coupler can be balanced during calibration. Each nominal 50 Ω load is realized with four 200 Ω resistors in parallel. The parallel network gives 50 Ω while distributing dissipation among four components. The current transformer uses an FT114-43 core with approximately 20 turns. These component choices support calibration.

The outputs of the couplers provide low-frequency analog voltages proportional to the forward and reflected RF power. These analog signals are then transferred to the control board, where they are digitized and further processed by the controller to estimate power levels and SWR.

4.2.6. Frequency measurement hardware

The RF board retains an optional frequency pickup and comparator circuit for each transceiver path. It was included to permit a later hardware-based frequency measurement that is independent of transceiver-specific CAT protocols. In the final integrated release, however, frequency counting is not implemented in software

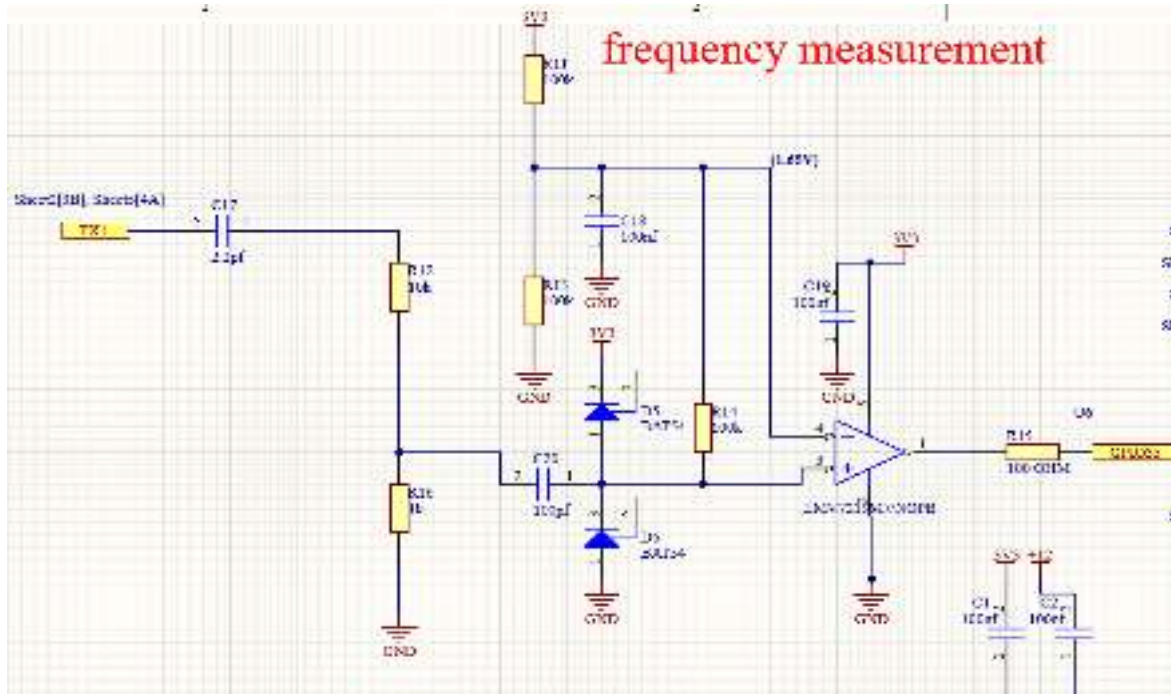


Figure 11 Frequency measurement system schematic

Each transceiver path initially included:

- A small RF pickup capacitor (C17),
- An attenuation and protection network (R12, R16 and C20),
- An LMV7219 high-speed comparator.

The RF pickup capacitor was intended to sample only a small fraction of the transmitted RF signal. This sampled signal was then attenuated and conditioned before being applied to the LMV7219 comparator. The comparator was intended to convert the analog RF waveform into a digital pulse train that could be counted by the ESP32 controller in order to determine the RF frequency.

During the first practical implementation, however, the frequency measurement function did not provide sufficiently stable and reliable results. The complete measurement chain depends on obtaining a properly conditioned RF sample, converting it into clean digital transitions and then counting these pulses correctly over the complete HF operating range.

In addition, the CYD platform introduced further limitations because only a limited number of suitable GPIO pins were available. Some of the available pins are input-only or already allocated to the display, touchscreen and other peripherals. The pulse-counting implementation on the ESP32 also added additional complexity when measuring frequencies across the HF range.

Because frequency measurement was an optional function and was not required for the primary antenna-switching and protection functionality, it was decided not to include it in the final implementation. The comparator-based frequency measurement circuit and the corresponding software were therefore removed from the final design.

Removing this functionality simplified both the hardware and firmware and allowed development to focus on the essential functions of the system: antenna switching, interlocking, forward and reflected power measurement, SWR calculation, local touchscreen control and remote Blynk control.

The frequency measurement circuit remains a possible future extension. A future implementation could use a dedicated RF prescaler or frequency-counter IC before the controller in order to provide a more reliable digital signal over the complete HF frequency range.

4.2.7. Control board

The control board contains the digital and low-voltage hardware required to operate the system. The main elements are:

- A. CYD platform with ESP32,
- B. MCP23017 I/O expander,
- C. TBD62083A relay driver stages,
- D. ADS1115 analog input interface,
- E. The supply distribution circuitry.

A. CYD platform

The CYD is used as the main controller and local display platform. It provides the processing capability for relay control, display handling, remote communication, and measurement processing. It also serves as the software platform for the user interface and Blynk communication.

A limitation of the CYD is the relatively small number of easily accessible external I/O pins.



Figure 12 CYD board

The limited number of accessible CYD pins required shared I²C peripherals and careful pin allocation. Calling **Wire.begin()** with the default ESP32 pins drove GPIO21 as SDA and caused the display backlight to flicker.

Initial 4.7 k Ω pull-ups also produced slow edges and intermittent acknowledgements on the loaded ribbon-cable bus.

The as-built solution uses **Wire.begin(27, 22)**, a 100 kHz bus, and 1 k Ω pull-ups to 3.3 V.

The lower resistance gives approximately 4.7 times faster RC rise time and about 3.3 mA LOW-state current.

B. MCP23017 I/O expander

Because the number of relay control signals exceeds the directly available GPIO resources, the design uses the MCP23017 I²C I/O expander. This provides additional digital outputs for relay control and allows the CYD to manage a large number of relays through only a small number of control lines.

Address pins A0, A1 and A2 are tied to ground, fixing the 7-bit address at 0x20 and preventing floating address inputs. The active-low RESET input is held at 3.3 V by R35 = 10 k Ω so that the device remains enabled during controller start-up or disconnection.

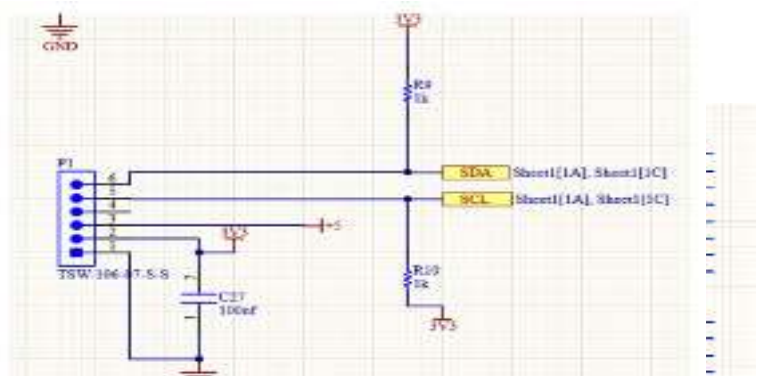


Figure 13 CYD interconnection and 1 k Ω I²C pull-ups.

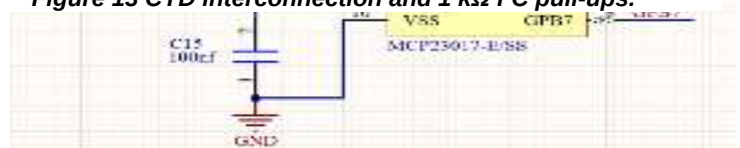


Figure 14 MCP23017 I/O expander

C. TBD62083AFWG relay drivers

Relay coils cannot be driven directly from the microcontroller or the MCP23017 I/O expander. These devices are intended for low-power digital signals and cannot supply the current required by the 12 V relay coils. Therefore, an interface stage is required between the 3.3 V control electronics and the relay supply.

In the first version of the design, ULN2803 Darlington transistor arrays were selected for this purpose. Each ULN2803 contains eight low-side driver channels and integrated suppression diodes for inductive loads, making it suitable for controlling several relay coils from digital logic signals.

During the redesign, the ULN2803 was replaced by the Toshiba TBD62083A due to component availability and lifecycle considerations. The TBD62083A provides the same basic function as an eight-channel low-side relay driver but uses DMOS output transistors instead of Darlington transistor pairs.

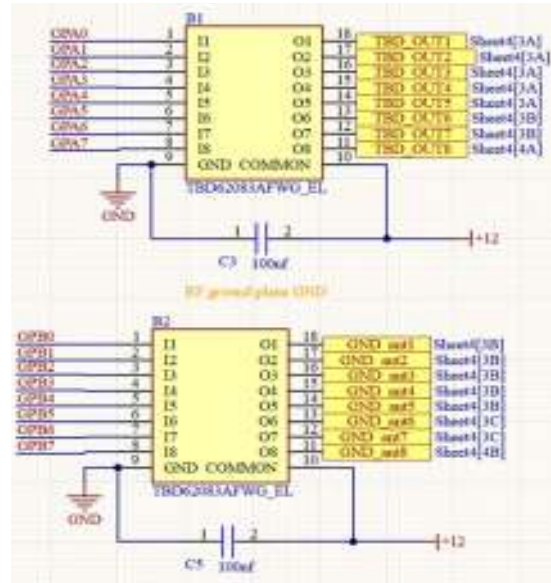


Figure 15 TBD62083AFWG relay drivers

The use of DMOS outputs provides lower conduction losses compared with the Darlington-based solution, reducing the voltage drop across the driver and the power dissipated inside the device. The TBD62083A is therefore a more suitable modern replacement for the relay-driving stage while maintaining the required eight-channel architecture.

TBD62083A also contains integrated clamp diodes for inductive loads. When a relay coil is switched off, the collapsing magnetic field generates a high reverse voltage, commonly referred to as an inductive kickback or flyback voltage. The internal clamp diodes provide a safe current path for this energy and protect the driver and the surrounding control electronics from the resulting voltage transient.

The TBD62083A therefore performs three main functions in this design:

- It provides the current required to energize the relay coils.
- It acts as a low-side interface between the 3.3 V control logic and the 12 V relay supply.
- It protects the control circuitry from inductive voltage spikes generated when the relay coils are switched off.

The change from the ULN2803 to the TBD62083A also makes the redesigned board less dependent on an older Darlington-based component while providing a lower-loss DMOS-based relay-driving solution.

D. ADS1115 analog interface

The analog outputs of the Bruene couplers cannot be processed directly by the digital control logic. Therefore, an ADS1115 analog-to-digital converter (ADC) is used to convert the measured analog voltages into digital values that can be read by the controller.

The ADS1115 was selected because it provides four analog input channels, which is exactly sufficient for the four measurement signals required in this project:

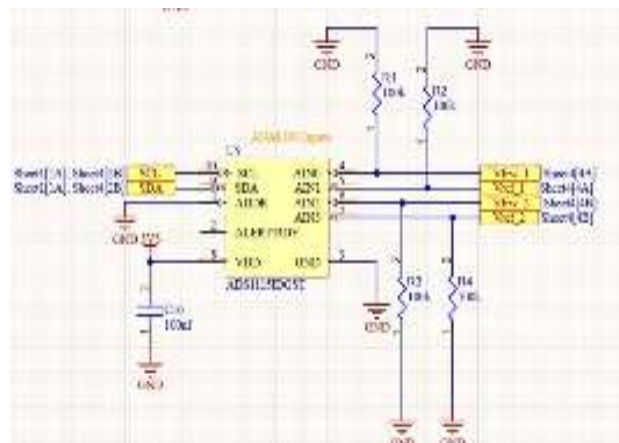


Figure 16 ADS1115 analog interface

- Forward signal for transceiver path 1.
- Reflected signal for transceiver path 1.
- Forward signal for transceiver path 2.
- Reflected signal for transceiver path 2.

The ADS1115 communicates with the controller through the I²C interface. This allows all four analog channels to be measured while requiring only the SDA and SCL communication lines, which is useful because the number of available GPIO pins on the CYD controller is limited.

The Bruene couplers produce detected low frequency/DC voltages representing the forward and reflected RF signals. These voltages are applied to the four single-ended ADS1115 inputs, where they are digitized and transferred to the controller. The measured values are subsequently used in software to estimate transmitted power and calculate the standing-wave ratio (SWR).

Each analog input also includes a 100 k Ω pull-down resistor to ground. These resistors were added after testing showed that the ADC input nodes could retain charge or behave as high-impedance floating nodes when no useful detector signal was present. This caused non-zero and unstable ADC readings even when no RF signal was applied.

The pull-down resistors provide each ADC input with a defined reference to 0 V when the detector output is inactive. They also provide a discharge path for the filtering capacitors connected to the measurement lines, allowing the measured voltage to return reliably to zero after the RF signal disappears.

A relatively high resistance of 100 k Ω was selected so that the pull-down provides a defined DC state without significantly loading the Bruene detector output. A much lower resistance would draw more current from the detector and could reduce the measured voltage.

The analog interface therefore performs both signal conversion and input stabilization. The ADS1115 provides accurate digital measurements of the four detector signals, while the 100 k Ω pull-down resistors prevent floating inputs and improve the stability of the zero-signal measurements.

E. Power supply hardware

The system is powered from an external 12 V DC supply. A TSR 1-2450E step-down module generates 5 V for the CYD, and a TSR 1-2433E module generates 3.3 V for the MCP23017, ADS1115, pull-ups, and other low-voltage circuitry. Each regulator has 100 nF input and output decoupling. The two switching-regulator circuits are shown in Figures 16 and 23.

Separate regulated rails avoid powering the CYD directly from the 3.3 V logic rail and reduce dissipation compared with linear regulation. The modules are rated as 1 A step-down regulators; the actual system load and thermal margin must still be checked under worst-case operation. The power architecture therefore includes:

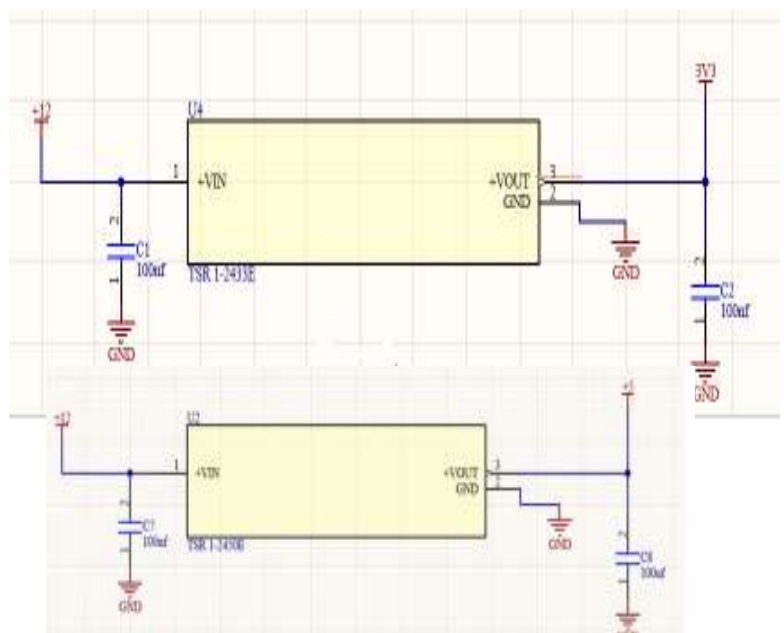


Figure 19 TSR 1-2450E 12 V-to-5 V switching regulator.

- A main 12 V input and relay-coil rail,
- A regulated 5 V rail for the CYD,
- A regulated 3.3 V rail for logic, ADC and I²C, plus 12 V distribution to the relay drivers.

Relay current paths were separated from sensitive analog and digital sections, especially on the control board. This is important because the relay coils draw switching current and can introduce noise into the supply network. By separating these current paths and keeping the logic supply properly regulated, the stability of the control and measurement circuits is improved

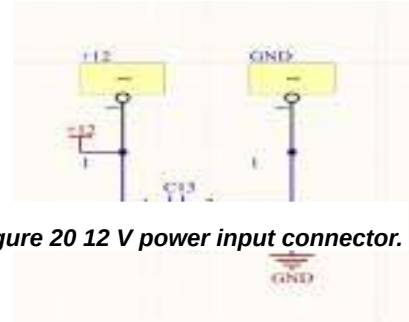


Figure 20 12 V power input connector.

4.2.8. Board separation and shielding

Because the project combines high-power RF switching with digital control electronics, physical and electrical separation between the RF section and the control section is an important part of the hardware architecture. The RF board and control board are implemented as separate PCBs in order to reduce unwanted coupling between high-power RF signals, relay switching signals and sensitive digital or analog circuitry.

This separation improves measurement stability, simplifies routing and makes shielding inside the metal enclosure easier to implement.

4.2.8.1. RF board

The first RF prototype was implemented as a 4-layer PCB. The intention was to use the top layer for the RF traces and an internal layer directly underneath as a continuous ground reference, while the remaining layers were used for relay power and supporting connections.

Although this structure provided a well-defined ground reference, the small dielectric spacing between the top RF layer and the internal ground plane required relatively narrow 50 Ω microstrip traces. After correction of the original stack-up calculation, the required RF trace width was approximately 0.77 mm.

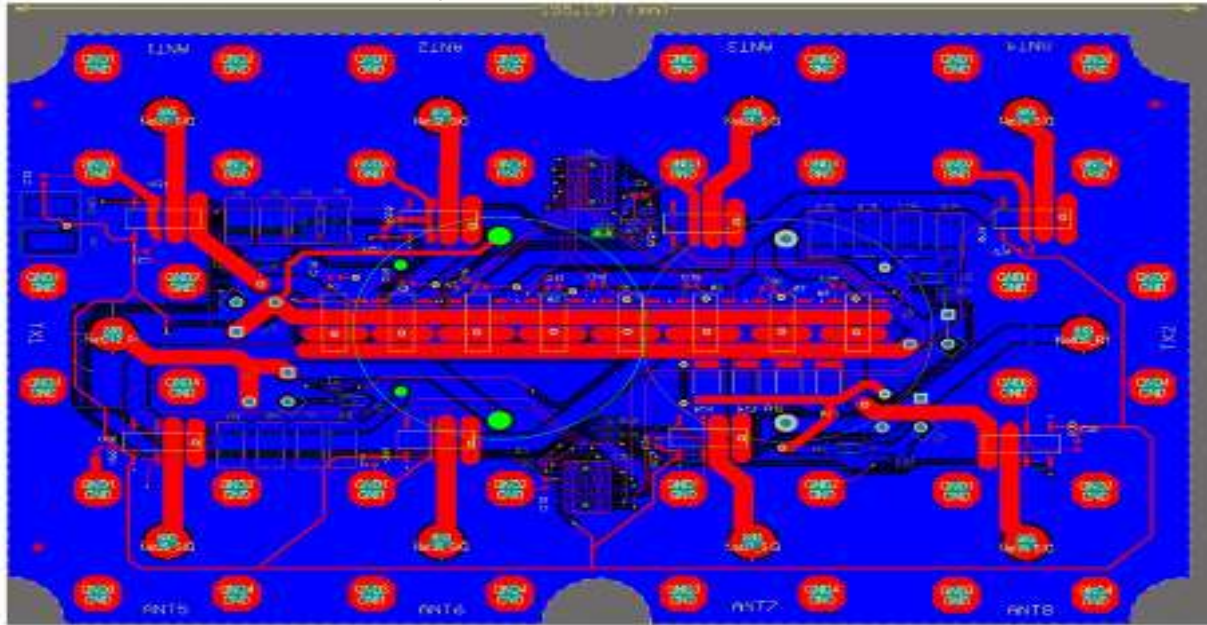
Practical VNA measurements showed that the first RF board was electrically functional, but its RF performance was below the intended target. At approximately 14.5 MHz, the prototype measured approximately -10.63 dB S₁₁ return loss and -0.46 dB insertion loss, while isolation and crosstalk were approximately -28.43 dB and -30.46 dB respectively. For this reason, the RF board was redesigned as a 2-layer PCB.

In the redesigned RF board, the top layer is used for the main RF transmission paths, relay connections and Bruene coupler circuitry. The bottom layer is maintained as a continuous ground plane underneath the RF traces.

Using a standard 1.6 mm 2-layer FR-4 construction results in a significantly wider 50 Ω microstrip trace, approximately 2.7–2.8 mm depending on the exact manufacturing stack-up. The wider traces are more suitable for the intended high-power RF application because they reduce conductor resistance, current density and thermal losses.

The redesigned RF board therefore focuses on:

- Short and direct RF paths,
- Approximately 50 Ω microstrip routing,
- Wider high-power RF conductors,
- A continuous bottom-layer ground plane,
- Strong grounding around the N-connectors,
- Short relay and connector transitions,
- Bruene coupler integration,
- Reduced insertion loss and Improved return loss and isolation.



4.2.8.2. Control board *Figure 21 RF-board routing, port arrangement and dimensions*

The control board contains the main low-voltage control functions, including:

- Digital control logic,
- MCP23017 I/O expansion,
- Relay-driver circuitry,
- ADS1115 analog acquisition,
- Communication with the CYD controller,
- User-interface support,
- A continuous ground reference for improved noise control.

By separating the RF hardware from the digital control electronics, the system becomes easier to route, debug and shield.

The physical separation also reduces the possibility that relay switching, digital communication or display activity couples into the RF paths or the sensitive measurement circuitry.

The final board architecture therefore combines a dedicated 2-layer high-power RF board with a separate 2-layer control board, each optimized for its own electrical requirements.

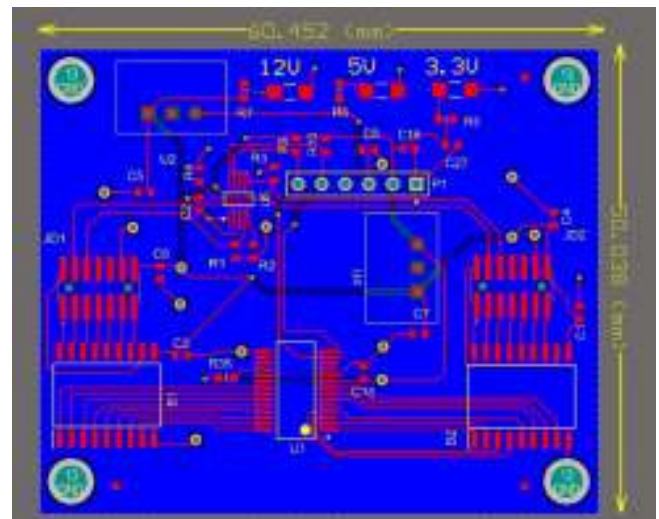


Figure 22 Control board layout

4.2.9. Enclosure and mechanical integration

For the enclosure, a die-cast aluminum housing is used (**Figure 20**). This type of enclosure was selected because the project combines high-power RF circuitry and digital control electronics, which makes mechanical robustness and electromagnetic shielding important design considerations.

A metal enclosure is preferred because it provides much better shielding than plastic housing. This helps reduce unwanted RF coupling, limits external interference, and improves the separation between the RF section and the control electronics. In addition, a die-cast aluminum enclosure is mechanically strong and well suited for mounting RF connectors and supporting the complete prototype in a practical way.

The enclosure will be machined to create the required openings for the N-type RF connectors.



These openings are planned to be made using a CNC machine, which allows accurate positioning and repeatable hole placement on the enclosure walls. This is important because the RF connectors must be mounted with sufficient mechanical precision and spacing to support both the external cabling and the internal RF interconnections.

4.2.10. Conclusion of the hardware design

The hardware design of this project is based on the integration of high-power RF switching and embedded control electronics into a compact two-board system. The design includes a relay-based switching matrix, ground-clamp functionality, directional couplers for power and SWR monitoring, hardware-based frequency detection, and a CYD-based control platform with expanded I/O and analog acquisition.

A major focus of the hardware design was the balance between practical implementation and RF performance. This required careful relay selection, short RF paths, strong grounding, appropriate measurement circuitry, and clear separation between RF and control electronics. The resulting hardware architecture provides a realistic basis for the realization and testing of the final prototype.

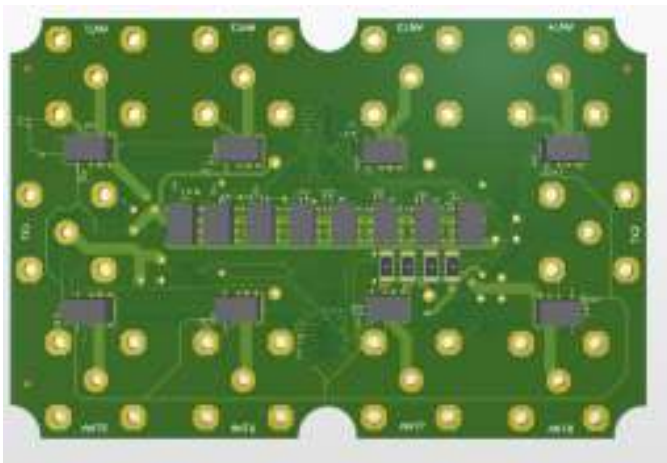


Figure 23 RF board 3D layout

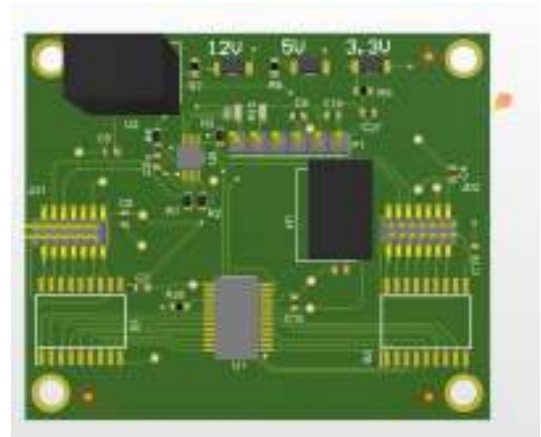


Figure 24 Control board 3D layout

4.3. Software design

The final firmware is implemented in the ESP32-based CYD. It combines relay control, detector acquisition, touchscreen input, Wi-Fi/Blynk communication, serial diagnostics, display updates, and watchdog supervision in one cooperative, non-blocking loop.

The loop services the touchscreen and network continuously, samples all four ADS1115 channels every 500 ms, and publishes Blynk values every second. Wi-Fi reconnection is attempted every 10 s, Blynk reconnection every 5 s, and each cloud connection attempt is bounded to avoid freezing local operation.

4.3.1. Software design general considerations

The implementation is divided into the following coordinated functions:

- Central antenna-state requests for local, remote and serial commands;
- MCP23017 relay-output control;
- ADS1115 acquisition, filtering and validity checks;
- Low-flicker graphical user interface and touch debouncing;
- Wi-Fi/Blynk synchronization and manual transmitter locks;
- A 12 s task watchdog and explicit start-up initialization.

No blocking delay is used in the main task except the short relay-transition delays.

This keeps touch, display, measurement, and reconnection functions responsive during normal operation.

4.3.2. Relay control

The MCP23017 is initialized at address 0x20. Port A outputs GPA0–GPA7 control the eight selector relays, and Port B outputs GPB0–GPB7 control the eight ground-clamp relays through the TBD62083AFWG stages. If the expander does not acknowledge during start-up, the firmware reports the fault and does not pretend that switching is available.

All user actions pass through *requestAntennaState()*. This single entry point applies occupancy rules consistently to touchscreen, Blynk and serial requests. When a requested bus is already occupied, the

previous antenna is parked before the new assignment is accepted, so each bus has at most one software owner.

4.3.3. State-based switching

Every antenna is represented by one of four enumerated states:

- ANT_BUS1 — connected to transceiver bus 1;
- ANT_BUS2 — connected to transceiver bus 2;
- ANT_PARK — released from both buses and grounded through the clamp relay;
- ANT_OFF — reserved for device-failure handling rather than normal operation.

After successful controller and expander initialization, all eight channels are commanded to ANT_PARK. The GUI maintains explicit BUS1 and BUS2 occupant fields so that the local display and Blynk interface reflect the same state model.

4.3.4. Safety and interlock logic

Two manual flags, *tx1Locked* and *tx2Locked*, reject connect or remove operations on a locked bus. They provide an operator-controlled interlock through the CYD and Blynk interface. They are not automatic PTT or RF detectors; therefore the operator must set the appropriate lock before transmission.

The hardware clamp is fail-safe when its coil is de-energized, but the current *connectAntennaHardware()* sequence releases the clamp, waits 25 ms, and then changes the selector. During that interval the antenna can briefly see the selector's previous bus. True break-before-make operation should set the selector while the antenna remains clamped, wait for contact settling, and only then release the clamp. This firmware limitation must be corrected before high-power switching is permitted.

4.3.5. Power and SWR processing

The ADS1115 is configured for *GAIN_ONE* and 475 samples/s. After every multiplexer change, one dummy conversion is discarded; four subsequent conversions are averaged. A first-order filter with $\alpha = 0.30$ reduces display noise, and detector voltages below 10 mV are forced to zero.

Displayed power is computed as $P = K \cdot V_{\text{forward}}^2$. The current constants $K1 = K2 = 10.0$ are provisional software values and must be replaced by calibration coefficients obtained with a known RF power standard. The software does not calculate peak-envelope power.

SWR is displayed only when forward voltage is at least 0.150 V, reflected voltage is below forward voltage, and the forward-minus-reflected difference exceeds 0.050 V. Otherwise the field shows "---". This prevents numerically unstable no-signal values from being presented as valid SWR, but it does not establish measurement accuracy.

4.3.6. Local user interface

The CYD interface shows eight antenna buttons, BUS1 and BUS2 selection buttons, PARK control, current bus occupants, two manual lock icons, connection status, forward and reflected detector voltages, calculated power, and conditional SWR. Static graphics are drawn once; changed values are redrawn selectively, with an off-screen sprite used where memory permits to reduce flicker.

Touch input is sampled three times, requires a pressure value above 90, and is latched so that one press generates one action. The display backlight remains on GPIO21 and is held HIGH after the I²C pins are explicitly initialized, eliminating the earlier backlight flicker.

4.3.7. Remote communication

Blynk mirrors the central state machine rather than controlling relays directly. V0 selects the active antenna; V1, V2 and V3 request BUS1, BUS2 and PARK; V4 and V5 report the bus occupants; V6–V9 publish the two power/SWR pairs; and V10/V11 control the manual transmitter locks.

Local touch, Blynk and serial commands receive the same validation. Loss of Wi-Fi or Blynk does not stop local operation because reconnection is periodic and bounded. Network credentials in the submitted source are placeholders rather than personal secrets.

4.3.8. Watchdog and fault tolerance

A 12 s task watchdog supervises the main loop. Start-up initializes the I²C bus, MCP23017, ADS1115, display and network services in a defined order, and the relay-state table is initialized to PARK after the expander becomes available.

A watchdog reset does not electrically reset the MCP23017, so expander outputs may persist until firmware initialization completes. The final design also lacks run-time I²C bus recovery and relay feedback. These limitations should be addressed with an expander reset connection or external power-on reset, periodic bus recovery, and position or current feedback where justified.

4.3.9. Conclusion

The final firmware demonstrates coordinated 8-to-2 state control, local and remote operation, detector acquisition, no-flicker display updates, and fault supervision. It is substantially beyond the original prospective design chapter and was exercised on the assembled prototype.

Nevertheless, manual locks are not automatic hot-switch prevention, the connect sequence requires a break-before-make correction, frequency counting is absent, and the power/SWR values are not calibrated. These boundaries are carried into the validation and discussion chapters.

5. Results

5.1. Final prototype realization

This chapter presents the implementation and bench-level validation of the final two-layer Smart Antenna prototype. Verified control and acquisition functions are distinguished from the expected RF-performance ranges that still require calibrated measurement.

5.1.1. RF-board implementation

The final RF board was implemented as a two-layer PCB using approximately 1.6 mm FR-4. The RF paths, relays, and directional couplers are located on the top layer, while the bottom layer forms a continuous ground-reference plane.

Based on the final PCB stack-up, a microstrip width of 2.7 mm was calculated in AppCAD to provide a characteristic impedance of approximately $50.52\ \Omega$ at 30 MHz. The RF routes were kept as short as practical, and the connector ground connections were placed close to the ground plane to reduce parasitic inductance and impedance discontinuities.

The board contains eight antenna inputs, two RF selector buses, ground-clamp relays, and two Bruene directional couplers. IM06DGR 12 V DPDT relays were used for both antenna selection and ground clamping. Both poles of each relay were connected in parallel to reduce contact resistance and improve RF-current sharing.

Each antenna is connected to the common contact of its ground-clamp relay. The normally closed contact is connected to ground, while the normally open contact connects the antenna to the selector stage. Consequently, loss of relay-coil power places the antenna in a defined grounded state.

5.1.2. Control-board implementation

The control board contains the CYD/ESP32 controller, MCP23017 I/O expander, ADS1115 ADC, two TBD62083AFWG relay-driver arrays, and the required power and interface circuitry.

The MCP23017 address inputs A0, A1, and A2 were connected to ground, resulting in the fixed I²C address 0x20. Grounding these inputs prevents an undefined device address caused by floating logic levels. The active-low RESET input was held high using a pull-up resistor so that the expander remains enabled during normal operation.

The ADS1115 was configured at address 0x48. Its inputs were assigned as follows:

ADC input	Signal
AIN0	TX1 forward-detector voltage
AIN1	TX1 reflected-detector voltage

AIN2	TX2 forward-detector voltage
AIN3	TX2 reflected-detector voltage

Four 100 k Ω pull-down resistors were added to the ADC inputs. These resistors provide a defined no-signal state and a discharge path for the detector-filter capacitors while only lightly loading the detector outputs. The final I²C interface uses GPIO27 for SDA and GPIO22 for SCL. GPIO21 remains assigned to the CYD display backlight. Reliable I²C communication was obtained using 1 k Ω pull-up resistors to 3.3 V. The relay coils are controlled through two TBD62083AFWG low-side DMOS driver arrays.

5.1.3. Firmware implementation

The final firmware integrates:

- antenna assignment and relay-state control;
- BUS1 and BUS2 occupancy management;
- grounded antenna parking;
- manual TX1 and TX2 locks;
- ADS1115 acquisition;
- provisional power and SWR processing;
- touchscreen operation;
- Blynk remote control;
- serial diagnostics;
- Wi-Fi reconnection; and
- watchdog handling.

Touchscreen, serial, and Blynk commands are processed through the same state-management function. Remote commands therefore cannot bypass the safety restrictions applied to local commands.

5.2. Bench-level validation

The final prototype was tested incrementally. Power and communication were verified first, followed by relay control, antenna-state management, touchscreen operation, Blynk communication, and analog acquisition.

Test	Result	Status
Supply rails	The 12 V, 5 V, and 3.3 V rails were present at their intended levels	Verified
MCP23017 communication	Device detected at address 0x20	Verified
ADS1115 communication	Device detected at address 0x48	Verified
I²C reliability	Stable communication obtained after replacing 4.7 k Ω pull-ups with 1 k Ω	Verified
Relay drivers	Both TBD62083AFWG arrays operated their respective relay coils	Verified
BUS1 and BUS2 selection	Antennas could be assigned to either bus	Verified
Bus occupancy	A maximum of one antenna was permitted on each bus	Verified
Conflicting assignment	One antenna could not be	Verified

	assigned to both buses simultaneously	
PARK state	Unused antennas were commanded to the grounded state	Verified
Start-up state	All antennas were initialized in the PARK state	Verified
Manual TX locks	Prohibited switching commands were rejected	Verified
Touchscreen control	Local selection and status indication operated correctly	Verified
Blynk control	Remote commands and status synchronization operated correctly	Verified
ADC acquisition	All four detector channels were acquired independently	Verified
No-signal ADC stability	The 100 k Ω pull-down resistors reduced residual inactive-input readings	Verified
Watchdog operation	Basic firmware recovery was demonstrated	Verified
Calibrated power and SWR	Absolute measurement accuracy was not established	Verified

The tests demonstrate correct bench-level operation of the control architecture. They do not prove safe switching while RF power is present. The transmitter locks are manually controlled, and the prototype does not include automatic PTT interlocking or relay-contact position feedback.

- Correction detection of the channel number,
- Consistency of the measurement results.

Once these points have been confirmed, the next step will be the calibration of the measurement chain in order to obtain meaningful power and SWR values. Because of the practical nature of the Bruene coupler implementation, the final behavior of the measurement subsystem must be verified experimentally.

All further updates related to power reading, SWR estimation, and measurement calibration will be documented in the GitHub repository.

5.3. RF measurement results

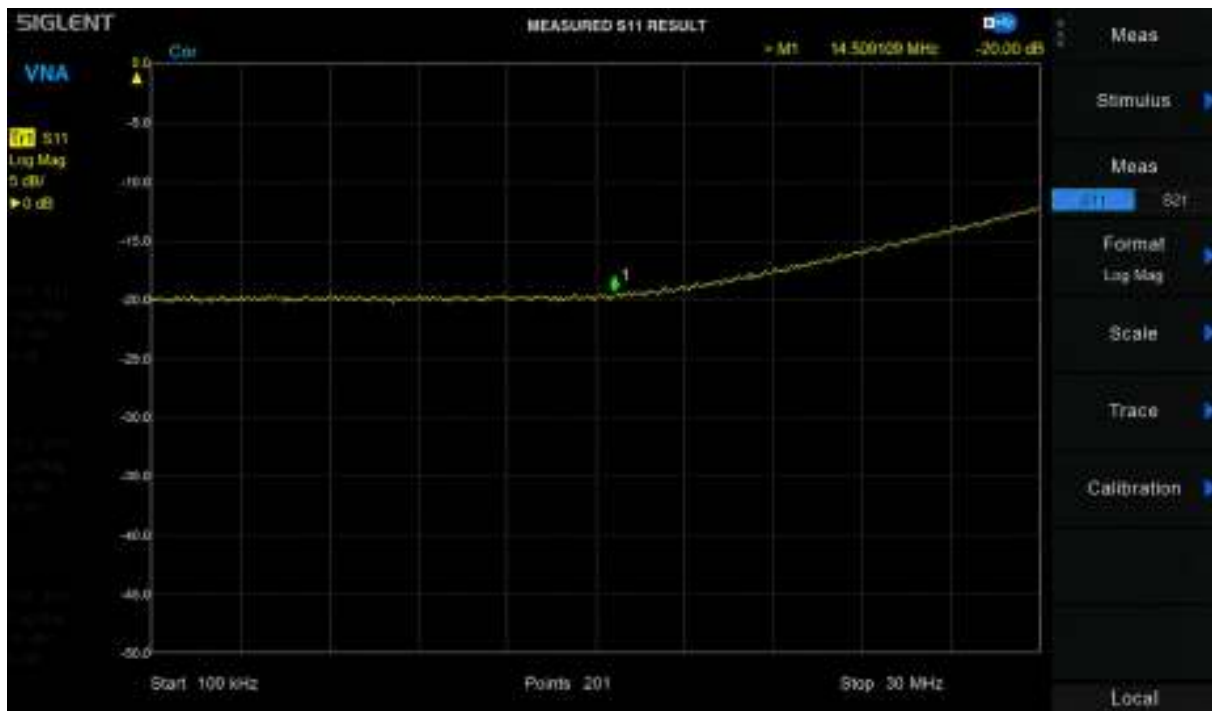


Figure 25 S11 measurement

Following bench-level validation, an initial low-power VNA measurement was performed on the assembled final RF board over the HF range. The selected RF path was terminated in 50 ohm, and the measured return loss was approximately 20 dB at frequencies below 15 MHz. This corresponds to S11 of approximately -20 dB, a reflection-coefficient magnitude of approximately 0.10, and VSWR of approximately 1.22. Under the measured conditions, approximately 1% of the incident power was reflected. The low-frequency response is therefore considered acceptable for bench-level evaluation of the prototype.

Toward the upper end of the band, the return loss decreased to approximately 13 dB at 30 MHz, corresponding to S11 of approximately -13 dB and VSWR of approximately 1.58. The RF path remained functional, although the result shows that the impedance match becomes less favorable as frequency increases.

No single physical cause is assigned from the magnitude measurement alone. Possible contributors include relay and contact parasitics, open relay branches, connector transitions, the directional-coupler region, and PCB discontinuities. The behavior will be investigated using repeated measurements and comparisons between RF paths. Revised S11 measurements and any resulting hardware improvements will be documented in the project GitHub repository: <https://github.com/voltversa/Intelligent-Remote-Controlled-Antenna-Matrix-for-Shortwave-Radio>

5.4. Power and SWR measurement status

Bench testing confirmed that all four ADS1115 detector channels can be acquired independently and that the corresponding forward and reflected detector signals can be processed by the controller. This verifies the operation of the analog acquisition chain at a functional level.

The displayed forward power, reflected power, and SWR values have not yet been calibrated against an RF power reference. They are therefore used as relative monitoring values at the present stage. Calibration data, revised coefficients, and updated measurement results will be added to the project GitHub repository as the characterization continues.

5.5. Safety validation

Bench-level testing confirmed that the control software permits only one antenna on each bus, rejects assignment of one antenna to both buses, places unused antennas in the grounded PARK state, and synchronizes local and Blynk commands. Manual transmitter locks also prevent relay-state changes when the corresponding transmitter is marked as active.

These results demonstrate controlled operation of the prototype in the bench environment. They do not constitute qualifications for switching while RF power is present or for continuous 1 kW operation. Until high-power validation is completed, operation remains supervised and relay switching is performed only while both transmitters are inactive.

5.6. Conclusion of the results chapter

The final prototype achieved the intended bench-level control and monitoring functions, including relay actuation, antenna-state management, grounded parking, local and remote operation, and four-channel detector acquisition.

The measured return loss of approximately 20 dB below 15 MHz demonstrates acceptable low-frequency RF performance for the current prototype. The decrease to approximately 13 dB at 30 MHz identifies an area for continued RF optimization rather than invalidating the functional concept. Investigation of this frequency-dependent behavior will continue, and updated S11, insertion-loss, isolation, and calibrated power-measurement results will be published in the project GitHub repository.

6. Interpretation of results

The principal result of the project is the realization of a functioning 8-to-2 antenna-routing architecture. The prototype can control the relay matrix, maintain one software occupant per bus, park unused antennas in a grounded state, synchronize local and remote commands, acquire four detector channels, and recover through a basic watchdog mechanism.

The final hardware implementation also validates the selected system architecture. The two-layer RF board provides short, wide signal paths over a continuous ground plane, while the separate control board integrates the CYD controller, MCP23017 I/O expander, ADS1115 ADC, two TBD62083AFWG relay-driver arrays, and the associated power and interface circuitry.

The approximately 20 dB return loss measured below 15 MHz confirms that the assembled RF path provides an acceptable impedance match for low-power bench evaluation in the lower HF range. The progressive reduction to approximately 13 dB at 30 MHz indicates a frequency-dependent mismatch. Because several RF transitions are present in the measured path, further measurements are required before the dominant contribution can be identified.

The available evidence therefore supports the description of a functional engineering prototype with successful bench-level control and initial RF validation. Absolute wattmeter accuracy, complete path-to-path RF characterization, and continuous high-power operation remain outside the completed validation.

7. Discussion

The final design represents an engineering compromise between RF performance, component availability, cost, PCB area, and assembly practicality. Dedicated coaxial RF relays would offer more controlled impedance, but the IM06DGR relay solution provides a compact and economical implementation suitable for prototype development.

The approximately 20 dB return loss obtained below 15 MHz shows that the two-layer RF layout, relay routing, and connector arrangement provide a useful impedance match in the lower part of the operating range. This result is considered acceptable for the present bench-level prototype and confirms the practical feasibility of the selected architecture.

The gradual degradation toward 30 MHz demonstrates that calculated microstrip impedance alone is insufficient to describe the complete assembled network. Relay-contact capacitance and inductance, open branches at unselected paths, connector launches, coupler discontinuities, and assembly tolerances may all contribute. The measurement trend will therefore be followed using calibrated VNA measurements, Smith-chart analysis, and comparisons between relay states and signal paths.

The Bruene couplers and ADS1115 interface provide a practical basis for forward- and reflected-signal monitoring. Independent acquisition of all four detector channels was verified, while conversion to absolute power and SWR values still depends on experimental calibration of each measurement path.

The safety functions implemented in software reduce the probability of invalid antenna assignments and unintended switching. Nevertheless, manual transmitter locks do not replace automatic RF or PTT interlocking, and high-power operation must remain supervised until the required protection and validation are completed.

Overall, the project demonstrates a complete and operational antenna-matrix prototype and establishes a clear basis for continued RF optimization. New measurements, investigation of the upper-band mismatch, and relevant hardware or firmware revisions will be documented in the project GitHub repository.

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Annexes

The annexes contain supporting technical material that complements the main thesis text. These annexes provide additional detail on the design, implementation, and testing of the system without interrupting the readability of the main report.

Annex A – Full RF board schematic

Complete schematic of the RF board, including the relay matrix, ground-clamp relays, Bruene couplers, and frequency pickup circuits.

Annex B – Full control board schematic

Complete schematic of the control board, including the CYD-related circuitry, MCP23017, ULN2803, ADS1115, comparator connections, and power supply section.

Annex C – PCB layouts

Top, bottom, and additional relevant views of the RF and control PCB layouts.

Annex D – PCB stack-up and grounding

Overview of the RF board 4-layer stack-up and the grounding strategy of both boards.

Annex E – RF and relay calculations

Voltage, current, and mismatch calculations used for relay and PCB design justification.

Annex F – Bruene coupler calculations and component sizing

Supporting calculations and design considerations for the directional coupler implementation.

Annex G – Microstrip calculations

Trace-width calculations and assumptions used for the RF bus design.

Annex H – Frequency measurement circuit details

Additional information related to the RF pickup, attenuation, protection, and comparator stages.

Annex I – Software excerpts

Relevant software fragments used for communication, relay control, measurement, and subsystem testing.

Annex J – Test setup and measurements

Additional screenshots, measurements, and observations obtained during hardware and software validation.

Annex K – Enclosure and mechanical drawings

Mechanical drawings and machining plans for the Hammond enclosure and RF connector placement.

Annex L – Bill of materials

Overview of the main components used in the prototype



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